

scripts/01_reference_dataset_eda.py

161 lines

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1 """
2 01_reference_dataset_eda.py
3 =====
4 Exploratory analysis of the PROVIDED reference dataset
5 (`data/official_sample_dataset.csv`, the hackathon's
6 `Autonomous_Choke_Control_Simulated_Dataset.csv`).
7
8 Per the problem statement, this dataset is used for exactly one purpose:
9
10 "The reference dataset is intended only to demonstrate simulator
11 behavior. Students are expected to generate their own data using the
12 simulator and develop their control-oriented models from these
13 experiments."
14
15 So NOTHING in this project's model identification touches this file. The
16 dynamic model in `03_identify_model.py` is fitted exclusively to our own
17 step-test experiments (`02_step_tests.py`), as instructed. This script only:
18
19 1. characterises the reference data (structure, steps, steady states)
20 2. compares it against our physics-based stand-in simulator, honestly
21 documenting where the two differ
22
23 Outputs
24 -----
25 figures/reference_dataset.png the provided data, as trends
26 figures/reference_vs_standin.png steady-state comparison
27 data/reference_steady_states.csv extracted steady-state operating points
28 """
29
30 from __future__ import annotations
31
32 import sys
33 from pathlib import Path
34
35 sys.path.insert(0, str(Path(__file__).resolve().parents[1] / "src"))
36
37 import numpy as np
38 import pandas as pd
39 import matplotlib.pyplot as plt
40
41 from config import DATA_DIR, FIG_DIR
42 from well_simulator import WellSimulator
43 from plotting import (COLOR_Q, COLOR_WHP, COLOR_FLP, COLOR_BHP, COLOR_CHOKE,
44 style_axis, savefig)
45
46 REF_CSV = DATA_DIR / "official_sample_dataset.csv"
47
48
49 def load_reference() -> pd.DataFrame | None:
50     if not REF_CSV.exists():
51         print(f" NOTE: {REF_CSV.name} not present - skipping reference EDA.")
52         print(" (The rest of the pipeline does not depend on it.)")
53         return None
54     df = pd.read_csv(REF_CSV)
55     df = df.rename(columns={
56
57         "Time_hr": "time_h", "Choke_pct": "choke_pct", "OilRate_bbl_hr": "Q",
58         "WHP_psi": "WHP", "FLP_psi": "FLP", "BHP_psi": "BHP",
59     })
60     return df
61
62 def steady_states(df: pd.DataFrame) -> pd.DataFrame:
63     """Average the last few samples of each constant-choke hold."""
64     df = df.copy()
65     df["hold"] = (df["choke_pct"] != df["choke_pct"].shift()).cumsum()
66     rows = []
67     for _, seg in df.groupby("hold"):
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68 tail = seg.tail(5).mean(numeric_only=True)
69 rows.append({"choke_pct": seg["choke_pct"].iloc[0], "n_samples": len(seg),
70 "Q": tail["Q"], "WHP": tail["WHP"],
71 "FLP": tail["FLP"], "BHP": tail["BHP"]})
72 return pd.DataFrame(rows)
73
74
75 def plot_reference(df: pd.DataFrame) -> None:
76 fig, axes = plt.subplots(5, 1, figsize=(11, 11), sharex=True)
77 t = df["time_h"]
78 axes[0].plot(t, df["choke_pct"], color=COLOR_CHOKE, lw=2, drawstyle="steps-post")
79 style_axis(axes[0], title="Provided reference dataset - choke position", ylabel="Choke [%]")
80 for ax, col, c, unit in [
81 (axes[1], "Q", COLOR_Q, "bbl/hr"), (axes[2], "WHP", COLOR_WHP, "psi"),
82 (axes[3], "FLP", COLOR_FLP, "psi"), (axes[4], "BHP", COLOR_BHP, "psi"),
83 ]:
84 ax.plot(t, df[col], color=c, lw=1.8)
85 style_axis(ax, title=col, ylabel=f"{col} [{unit}]")
86 axes[-1].set_xlabel("Time [h]")
87 fig.suptitle("Reference dataset (illustrative only - NOT used for model identification)",
88 fontweight="bold")
89 fig.tight_layout()
90 savefig(fig, FIG_DIR / "reference_dataset.png")
91
92
93 def plot_comparison(ss: pd.DataFrame) -> None:
94 """Overlay the reference steady states on our stand-in's own curves."""
95 sim = WellSimulator(noise=False)
96 u_grid = np.linspace(20, 75, 120)
97 ours = pd.DataFrame([{"choke_pct": u, **sim.steady_state(u)} for u in u_grid])
98
99 fig, axes = plt.subplots(1, 4, figsize=(15, 3.9))
100 for ax, col, c, unit in [
101 (axes[0], "Q", COLOR_Q, "bbl/hr"), (axes[1], "WHP", COLOR_WHP, "psi"),
102 (axes[2], "FLP", COLOR_FLP, "psi"), (axes[3], "BHP", COLOR_BHP, "psi"),
103 ]:
104 ax.plot(ours["choke_pct"], ours[col], color=c, lw=2, label="our stand-in")
105 ax.plot(ss["choke_pct"], ss[col], "o--", color="#111827", lw=1.4,
106 ms=6, label="reference data")
107 style_axis(ax, title=col, xlabel="Choke [%]", ylabel=f"{col} [{unit}]")
108 ax.legend(fontsize=8)
109 fig.suptitle("Steady-state behaviour: physics stand-in vs provided reference dataset",
110 fontweight="bold")
111
112 fig.tight_layout()
113 savefig(fig, FIG_DIR / "reference_vs_standin.png")
114
115 def main():
116 df = load_reference()
117 if df is None:
118 return
119
120 print(f"Reference dataset: {len(df)} rows, choke levels "
121 f"{sorted(df['choke_pct'].unique().tolist())}")
122 ss = steady_states(df)
123 ss.to_csv(DATA_DIR / "reference_steady_states.csv", index=False)
124
125 print("\nSteady-state operating points extracted from the reference data:")
126 print(ss.round(1).to_string(index=False))
127
128 # Characterise its gain, for comparison with our own step tests.
129 ss_sorted = ss.sort_values("choke_pct")
130 du = np.diff(ss_sorted["choke_pct"].values)
131 dq = np.diff(ss_sorted["Q"].values)
132 gains = dq / du
133 print(f"\nReference dQ/du across its range: {np.round(gains, 2).tolist()} bbl/hr per %")
134 print(f" -> min {gains.min():.2f}, max {gains.max():.2f} "
135 f"(ratio {gains.max() / gains.min():.1f}x)")
136
137 sim = WellSimulator(noise=False)

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138 ours = [sim.steady_state(u)["Q"] for u in ss_sorted["choke_pct"]]
139 og = np.diff(ours) / du
140 print(f"Our stand-in over the same range: {np.round(og, 2).tolist()} bbl/hr per %")
141 print(f" -> min {og.min():.2f}, max {og.max():.2f} (ratio {og.max() / og.min():.1f}x)")
142
143 print("\nKEY DIFFERENCES (documented in the report, not corrected for):")
144 print(" * the reference is close to LINEAR in choke; our physics stand-in is")
145 print(" strongly nonlinear because of the choke orifice equation")
146 print(" * the reference FLP FALLS as rate rises; a flowline with friction")
147 print(" should see FLP RISE with throughput, as our stand-in does")
148 print(" * absolute pressure levels differ (reference WHP ~217-270 psi,")
149 print(" BHP ~2890-3130 psi)")
150 print("\nNOTE: this dataset is used for ILLUSTRATION ONLY. The dynamic model is")
151 print("identified exclusively from our own step tests, per the problem statement.")
152
153 plot_reference(df)
154 plot_comparison(ss)
155 print(f"\n wrote {FIG_DIR / 'reference_dataset.png'}")
156 print(f" wrote {FIG_DIR / 'reference_vs_standin.png'}")
157
158
159 if __name__ == "__main__":
160     main()
161

```